

CompTIA Server+

Server Hardware Maintenance

- Introduction
 - Lab Topology
 - Exercise 1 - Local Hardware Administration
 - Exercise 2 - Out-of-Band-Management
 - Exercise 3 - Hot-Swappable Hardware
 - Exercise 4 - BIOS and UEFI
 - Review
-

Introduction

Server+

KVM

Crash Cart

Serial and Parallel Connections

Remote Drive Access

Remote Console Access

Remote Power On/Off

Hot Swap

BIOS

UEFI

Firmware Updates

Welcome to the **Server Hardware Maintenance** Practice Lab. In this module, you will be provided with information about the maintenance of server hardware.

This module will introduce you to local hardware solutions and remote options. It will cover hot-swappable hardware and other hardware that helps to keep your servers constantly functioning. You will also talk about the software that allows the hardware to function, be configured, and boot up.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Local Hardware Administration
- Exercise 2 - Out-of-Band-Management
- Exercise 3 - Hot-Swappable Hardware
- Exercise 4 - BIOS and UEFI

After completing this module, you should have further knowledge of:

- Keyboard Video Mouse (KVM)
- Crash Cart
- Serial Connectivity Vs Parallel Connectivity
- Console Connection
- Virtual Machine Administration Console
- Remote Drive Access
- Remote Console Access
- Remote Power On/Off
- Internet Protocol Keyboard Video Mouse (IP KVM)
- Hard Drives
- Network Attached Storage Cages
- Network Interface Cards (NIC)
- Power Supplies
- Fans and Cooling
- BIOS
- UEFI
- Firmware Updates

Exam Objectives

The following exam objectives are covered in this module:

1.3 Given a scenario, perform server hardware maintenance

- Out-of-band management
- Local hardware administration
- Components
- Drives
- Hot-swappable hardware
- Basic input/output system (BIOS)/Unified Extensible Firmware Interface (UEFI)

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

This module contains supporting materials for Server+ (SK0-005).

Click **Next** to proceed to the first exercise.

Exercise 1 - Local Hardware Administration

In this exercise, local hardware that can help administrators be more efficient will be discussed. You will also learn about the KVM switch, a crash cart and its importance. Different management consoles that can help administrators troubleshoot and get systems back up and running, helping to ensure the Service License Agreements, will also be explained.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Keyboard Video Mouse (KVM)
- Crash Cart
- Serial Connectivity Vs Parallel Connectivity
- Console Connection
- Virtual Machine Administration Console

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Keyboard Video Mouse (KVM)

Keyboard Video Mouse (KVM) is a hardware for connecting several computers to a keyboard, display, and mouse using an ethernet or fiber optic connection. Users can switch between several computers connected with a simple touch of a button. These are mostly used in data centers and with servers because of their ease of use for administrators. The goal is to conserve space, resources, and money. Companies save money by not buying equipment for each computer and instead using the same connection to administer several computers. They reduce the amount of electricity utilized and hence save money. The amount of energy saved by not needing to have separate monitors for each computer is enormous. The amount of room required is also reduced. If every server and computer in a data center had a monitor, keyboard, and mouse, the space required to hold everything would be substantially larger.

Crash Cart

The term crash cart originated from the healthcare industry, providing a cart with basic medicines and bandages on a cart that could be transported to wherever needed. It was adopted in the IT industry for a cart used in a server room that has a laptop mounted to it and other useful tools. The cart will likely have a selection of cables to connect with the devices. Removable hard drives and backup media like tape drives should also be present. The cart should have enough room to work on and have ergonomics built in. A cart that can be easily adjusted for height is preferred. When a server needs to be serviced, the crash cart allows an administrator to easily connect to it and troubleshoot, update, and administer the server. These crash carts can help to get the equipment that is down working quickly, which helps with the Service License Agreements. Having servers down equates to lost money, so every minute matters.

Serial Connectivity Vs Parallel Connectivity

Serial Connection

A **Serial Connection** is established by sending data using one wire, which helps keep the network's cost down. It is because of this and the distance these networks can traverse that this technology is most used. Data will flow back and forth on that one wire in a full duplex one bit at a time and is not susceptible to timing issues. Serial connections require a converter as the computer cannot natively understand them. There are three types of Serial connections:

- **Asynchronous Transmission** - can be random and happen at any time
- **Synchronous Transmission** - this is continuous
- **Isochronous Transmission** - this is setup for transmissions at regular intervals

Parallel Connection

A **Parallel Connection** is when data is sent between two devices using multiple lines of communication. Think of a computer connected to another with five cables sending the data over it. The transmission is much faster with all those lanes of traffic opened compared to the one you get with a serial connection. There is also no conversion necessary as the computer hardware uses parallel communication as its underlying technology. The downfall comes in the ability to scale the network up. It becomes increasingly hard because the additional wires need to be managed. If the distance needed to travel is more than one hundred meters, repeaters must be implemented. The cost to create and manage these networks can quickly grow due to the excess amount of cabling and equipment.

Console Connections

Console connections refer to direct connections to Routers, Switches, Servers, and PCs in a network used for administrative purposes. Some devices, like routers and switches, do not have interfaces to work with, so direct connections through the console port are necessary to configure and make changes. Because this is a secondary connection, it can be used to help introduce devices to the network and act as redundancy to troubleshoot and repair in the event of a network failure. Common forms of connection are done through web pages or SSH. These are essential for administrators to do their jobs efficiently and effectively.

Virtual Machine Administration Console

The Virtual Machine Administration Console allows administrators to manage virtual machine tasks like installation, configuration, and troubleshooting. Virtual Machines are becoming increasingly more common, especially with cloud technology becoming more affordable and efficient. This helps organizations save money by either utilizing their own equipment and servers more efficiently or building a network in the cloud and saving on equipment purchase and upkeep costs.

Managing virtual machines can sometimes become difficult over time, especially in large organizations. VM Sprawl can occur when the number of Virtual Machines being managed gets to an amount the administrator can no longer manage effectively. This might be due to the fact that when a user leaves the organization, the account and machine are not disabled, or a hacker has created machines. The console allows for simplified administration and auditing. Using the

organization naming convention, they can use it to find users or virtual machines. Tools are built into the console to manage and give reports about the network and each machine. This can be done in the cloud or for the on-site network, wherever you have the VM server located. Windows has a built-in version for Microsoft Hyper-V, but there are versions for third-party Virtual Machine management too.

Exercise 2 - Out-of-Band-Management

Out-of-Band-Management is a technology that allows us to administer systems and networks securely from outside the LAN. A secure and dedicated secondary connection is established. This is extremely beneficial for managing systems when a network is unavailable, devices are in power-saving modes, or devices are entirely shut off. **Routers, switches, servers, power, and storage**, which are critical to our networks, may be accessible for troubleshooting and regular administration. Out-of-Band Management ensures that the Service License Agreement (SLA) guarantees are met. The goal is to ensure that administrators have constant access to important network components to address problems.

In this exercise, Remote Drive Access, Remote Console Access, Remote Power On/Off and Internet Protocol Keyboard Video and Mouse will be discussed.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Remote Drive Access
- Remote Console Access
- Remote Power On/Off
- Internet Protocol Keyboard Video Mouse (IP KVM)

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Remote Drive Access

Having access to drives remotely means you can access the data when needed. You may be accessing a PC with Remote Desktop Protocol, Network Attached Storage (NAS), Storage Area Network (SAN), or Cloud storage. Let us explore these technologies further.

Remote Desktop Protocol

Remote Desktop Protocol is a tool built into windows that allow users to access a computer from a different computer in a different location and work on it as if they were sitting in front of it. There are versions of RDP available for Mac and Linux. This protocol is used for productivity and administration and uses a default port of 3389 for the connection. Only one user can be logged into the PC at a time, so a user connected would be disconnected and could not interact with the session. If a technician were looking to help a user, remote assistance or quick assist would be better options. There are also third-party programs available that provide the same level of access. Each one is unique, designed to have some additional features built right in. Examples are Chrome Remote Desktop, Connectwise, TeamViewer, Zoho Assist, Take Control, Ninja One and FixMe.

Network Attached Storage (NAS)

Network Attached Storage (NAS) is a hardware and software solution for SOHO and medium-sized businesses. It is used for storing users' files and acts as a centralized file server through its software. The piece of hardware contains RAM, a Processor, Hard Drives for storage and a connection to the network. Users connect to, access, and work with the required data, which is then saved to the NAS. Due to the necessity of having data available, many NAS devices can employ RAID for fault tolerance and recovery of damaged drives. If a disk is damaged, you can replace it and utilize the RAID to rebuild the information.

To some extent, network linked storage can be scaled. If you run out of space, just add another drive to an empty bay or increase the capacity of the NAS's disks. This is thought to be a user-friendly technology that does not necessitate a lot of administration. Network Attached Storage does have some drawbacks. It cannot handle a large number of simultaneous connections. Some NAS software will cap the number of users connected at 25 to 30 hosts connected. Single Points of Failure can also exist with the hardware. For example, if the backplane that powers the disks fails, the NAS will be unavailable.



Figure 2.1: Displaying a Network Attached Storage (NAS) device.

Storage Area Network (SAN)

Storage Area Network (SAN) is a technology employed by large organizations to build a network to store data independently from the LAN. Because of the use of fiber optics, this technique is quite expensive. Fiber connections will be required to connect the SAN and client devices, which often already have an ethernet network. A Fibre Channel switch will be required, which is more expensive than a copper wire ethernet switch. In addition, the technology employed is more advanced than that of a NAS. Additional storage controllers, SAN servers, and other storage arrays will be used in a SAN. Because of the ability to scale up and establish full fault tolerance, technical expertise in setting up and administering this type of network will be necessary. However, the payoff is a significantly quicker and more efficient network with high availability and uptime.

SAN networks can transfer data at speeds of up to forty gigabits per second. SAN networks can support a large number of concurrent users. The number might be in the thousands, depending on the controller used. Instead of directly accessing the storage media like in the NAS, users send their requests to a server. The server sends those requests through the fibre channel and to the storage array. The data is retrieved and sent back through the fiber channel to the server and finally to the user. It may seem like a much longer and more complicated process, and it is, but with the blazing speeds, it is significantly faster than what a NAS can provide.



Figure 2.2: Displaying SAN Storage hard drives in a data center.

Cloud Storage

Cloud Storage has a couple of different types. There are Private Clouds that are owned and maintained by an organization. Public clouds utilize cloud storage providers resources to allow users to connect and store their data. The cloud storage operator will manage all the hardware and networking aspects. The purpose of cloud storage is to allow a user to connect, store, and access their data anywhere they have an internet connection.

Many cloud storage providers have applications for easy access and will also automate the process of updating the files once they have changed from a local source. Smartphone applications allow for access to these files anywhere there is a signal. Another purpose for cloud storage could be for backups and recovery. Instead of keeping these on-site, storing them on a server in the cloud could help in the instance of a natural disaster. However, there is a downside to this in the latency necessary to move the files to and from the cloud environment. Cloud storage also creates an environment for development. Developers and testers share drives and content to collaborate. Azure, AWS, and Google are three common public cloud service providers, but many others are available too.



Figure 2.3: Displaying Cloud Storage Technology.

Remote Console Access

Console Server

A Console Server helps to provide remote access to many devices in a centralized fashion. It is a tool that can help administrators in reaching their SLA goals. These servers are usually connected through a secondary connection. If the main network were to go down, the console server could still be accessed, and troubleshooting of the network could occur.

The console server hardware looks like an enterprise switch or router. These pieces of hardware can get up to ninety-six serial ports for making ethernet connections to the Servers, PCs, Routers, Switches, and other devices in the network. Once the connection is established, an administrator would have access to managing those devices as if they were sitting in front of it from a built-in management program or through SSH. These connections are secure and require the administrator to authenticate to the console server. This is an especially useful tool for troubleshooting our network and providing another remote way to administer the network and be more efficient. Console Servers are especially useful when there are network outages. The lab environment has console mode, which can be accessed if the normal connection is not functioning properly. The screenshot below shows how to access the console mode.

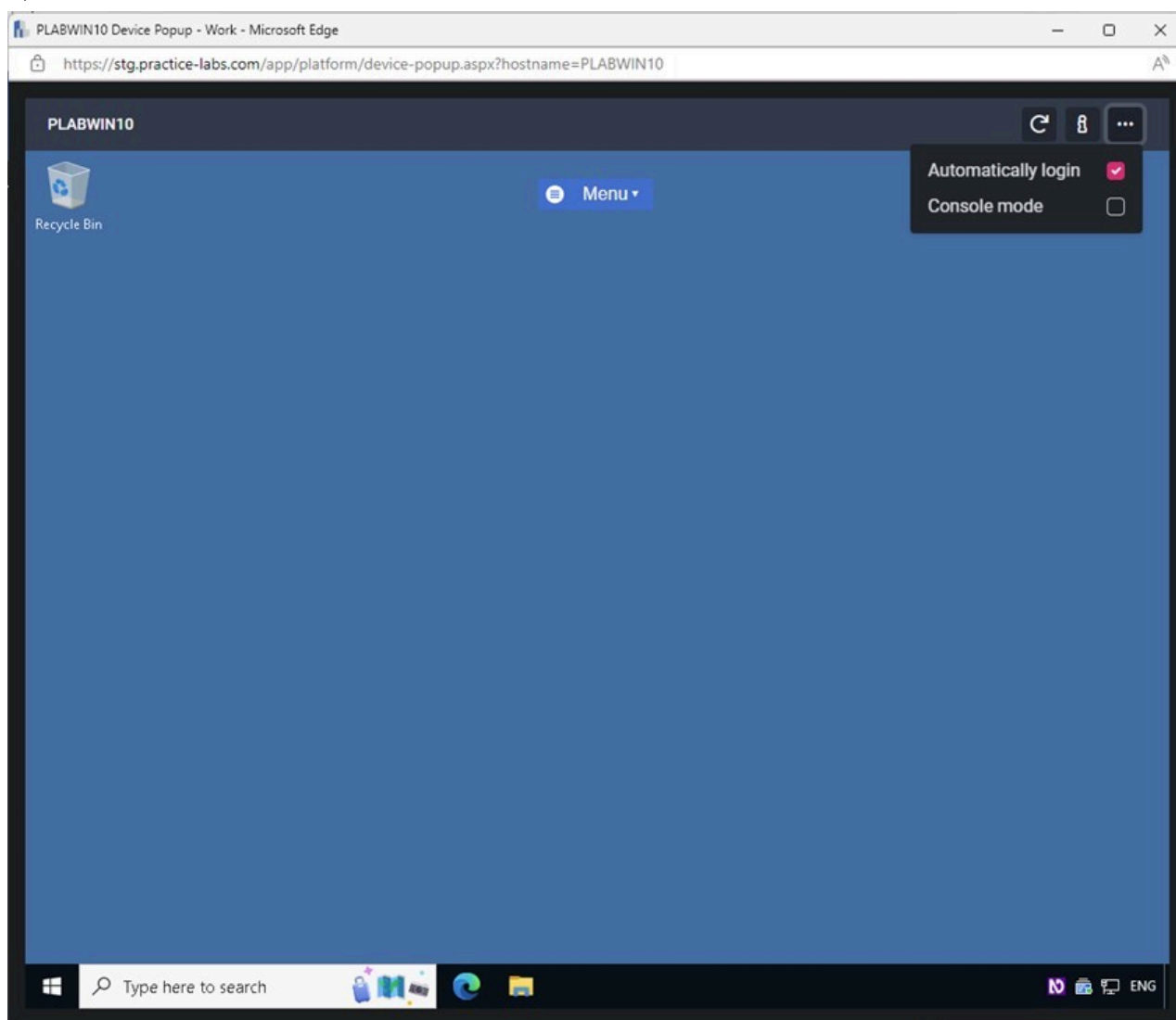


Figure 2.4: Displaying how the Console mode can be accessed in a lab device.

Remote Power On/Off

Wake-on-LAN

Sometimes it is not always feasible for an administrator or technician to physically access a computer system. With advancements in technology, remote access has become increasingly popular. In these instances, remotely powering the machine on and off from anywhere in the world with an internet connection is extremely useful. To do this, administrators will create a packet known as a magic packet or Wake-on-LAN packet using the scripting language. Once the packet is crafted, it is sent to the intended device on ports 7 and 9. This feature must be enabled through the Bios/UEFI menu and adapter settings.

The computer being woken up will be in a low power consumption state. If it is unplugged, be aware it will slowly drain the battery looking for a packet being sent to wake it. Not every computer supports this feature. Depending on the motherboard or network adapter, most do.

The below screenshot shows the device manager being accessed and the adapter's properties, showing the **Wake on Magic Packet Enabled**.

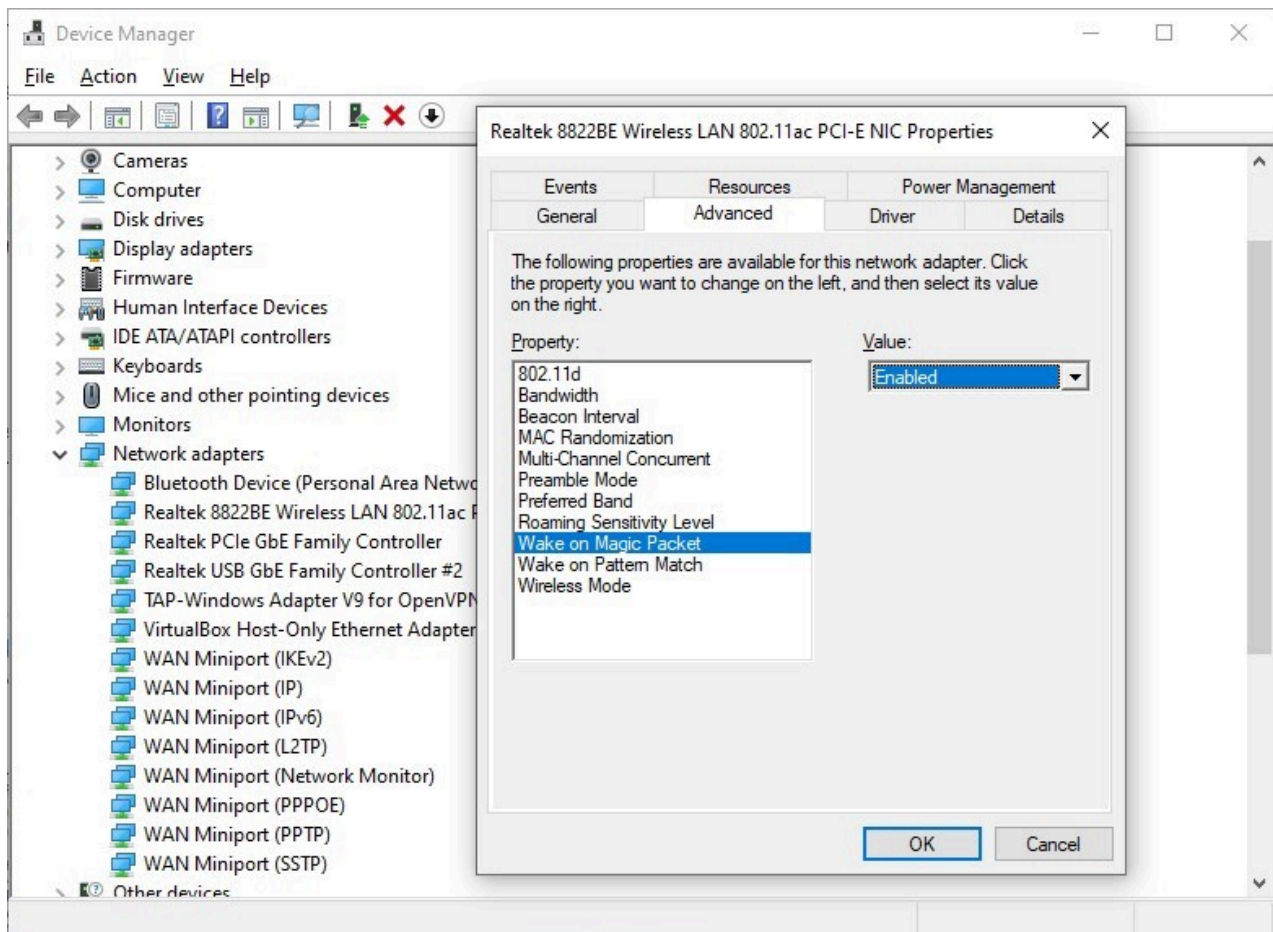


Figure 2.5: Displaying the Wake on Magic Packet Property enabled on the Adapter's Properties window.

Internet Protocol Keyboard Video Mouse (IP KVM)

Internet Protocol Keyboard Video Mouse (IP KVM) is a remote access tool that is a piece of hardware that allows a user to connect to a computer over an IP connection. This KVM plugs into the Keyboard, Video Monitor, and Mouse but also connects to the network with an IP address allowing users to access that IP address from anywhere in the world with a web browser or can be limited to in-band connections. Because the technology is not reliant on the host operating system to supply the interface it makes, it can be accessed from any operating system.

You will see the computer connected to the KVM instead of the screen of the host operating system. This connection, when configured correctly, is secure, fast, reliable, and easy to setup and use. Numerous monitors, multiple concurrent host access, USB access and control of USB devices, redundancy and fault tolerance power and internet connections, improved access, network load, and video quality are all possible with IP KVM. The ability to implement

Authentication servers increases the security already provided by HTTPS. This technology has a lot to offer, and prices will vary depending on the brand and size needed.

Exercise 3 - Hot-Swappable Hardware

There is a slight difference between Hot-swappable and Hot-pluggable. Hot-pluggable means equipment can be added to the PC with little interruption to the machine's operation. Hot-swappable replaces a part in the machine without shutting it down and having little to no interruption to the machine's operation. Every minute of downtime is crucial to avoid.

In this exercise, you will learn about hot-swappable hardware.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Hard Drives
- Network Attached Storage Cages
- Network Interface Cards (NIC)
- Power Supplies
- Fans and Cooling

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Hard Drives

Hot-swappable drives can be added without having to shut down the system. You can provide high amounts of uptime to hit the SLA agreements, and this technology helps to keep servers and other storage technologies running. HDD and SSD are both capable of doing this. Some forms of RAID, NAS and SAN storage will utilize this technology.

Hot-swappable drive technology is incredibly useful in manufacturing environments as well. Machines generate heat, dust, and vibration that can damage drives. Being able to hot-swap the broken drive for a new one saves time and money because production does not have to wait for the system to be shut down, the drive replaced, and the system back up again.

Some storage arrays will contain additional drives waiting on standby to be used if another drive fails. Swap the bad drive with a good one when the drive fails, and the standby drive becomes active. Most drives today have the capability to be hot-swappable. The most common issue is that sometimes they are not detected when plugged in and might need to be reconnected. Look for any extra space toward the back of the enclosure and ensure the bay it is plugged into is meant for that size drive. It is a quick and straightforward process that leads to constant service and zero downtime when deployed correctly.

A real-world example of this in action is when there is a RAID 1 in use. RAID 1 creates what is known as a mirror. One drive is duplicated exactly onto another drive. If one drive fails, the mirrored drive will take over, containing all the same data. You still need to replace that failed drive, though. With hot-swap, simply unplug the old drive and insert the new drive, and the RAID will rebuild without interrupting the service.



Figure 3.1: Displaying hot-swappable network RAID drives.

Network Attached Storage Cages

Cages refer to the form factor that the drives are added to in network attached storage. The cages will have their own management software or operating system, a CPU, Memory, a wired or wireless Network Interface Card, and the ability to add additional drives externally. The internal drives can be HDD or SSD, but HDD is much more common due to the lower cost. These drives are connected to a power rail in the back of the unit called a backplane. There are several single points of failure with these devices, and that backplane is common. If that backplane fails, there are no connections to the drives or data. If the Network Interface Card fails, there is no access to the drives or data. Suppose the CPU or Memory fails, there will be no access to the drives or data. So, although the unit itself can create redundancy with the data

on the disks, NAS often has no redundancy built into the hardware keeping the costs down for home users and small to medium size businesses. There are varied sizes and features available depending on the needs and budget of the purchaser.

Network Interface Cards (NIC)

Network Interface Cards allow computers and servers to talk with other machines through ethernet, fiber optic, and wireless connections. These devices work at layers 1 and 2 of the OSI model. Each network interface card has a unique identifier that is inscribed onto the piece of equipment. This is known as the device's physical address, mac address, or burned in address.

Desktop computers will have one NIC, but servers will likely have redundancy built in. Each server will likely have multiple network interface cards, which can all be active, providing additional avenues of traffic, or they might be setup in a standby setup, ready to take over if there is an issue with the other NIC.

Network interface cards are often hot-pluggable but may not be hot-swappable depending on the computer's operating system, BIOS, and motherboard. Servers will have the capability to hot-swap where the desktop computer may not.

When removing a network interface card, the following should be done:

- The card that is replaced on the PC will first need to be disabled so it does not communicate with other devices.
- Ensure ESD standards are met by using a mat or an ESD wrist strap and removing the case.
- Unplug the network card from its PCIe slot, plug in the new network card, and reassemble the case.

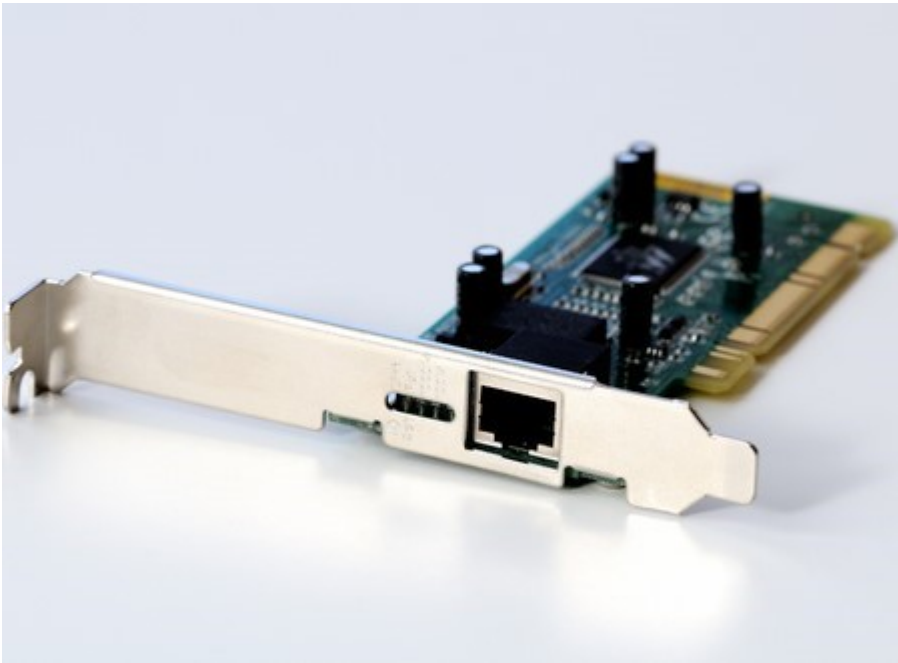


Figure 3.2: Displaying a Network Interface Cards (NIC).

Power Supplies

When it comes to servers and power supplies, there is a lot to consider. There are various levels: Titanium, Platinum, Gold, Silver, Bronze, and White. The more precious the metal, the higher the quality and efficiency should be. You want these power supplies to be efficient because, in the end, efficiency equals less energy and less money spent. Servers are expected to run 24 hours a day, 7 days a week. You will require the most efficient power supply if multiple servers are running.

Other considerations would be how much output is being generated, compatibility with the server hardware, and form factor would also be something to consider when choosing a power supply. It must fit the server and be able to power the server components with constant power. You do not want your servers shutting down without going through the proper process. There is the potential for too much data loss. As discussed, you want redundancy in most things involving servers.

You will not be looking for one power supply for the server but for an additional one too. Redundancy will be required for the power supply and where those power supplies plug into. Separate plugs with separate lines of power should be running to the plugs. This will ensure that if one is not active, the other can provide the power needed to the server. You will also include Uninterruptible Power Supplies (UPS) and Generators. A UPS is a battery backup that can help provide time to shut down the system gracefully or keep the machine running until the generator can take over when there is a power outage.



Figure 3.3: Displaying server power supplies.

Fans and Cooling

A huge challenge with Servers is making sure that they do not overheat. Proper setup of a server room can lower the room's temperature, the electricity needed, and the overall functioning of the server. The actual server case will have fans built into it that will move air over the components. The rack that the server is mounted in is designed to allow for proper airflow. One side will allow cool air to flow, and the other will push the hot air out. There are also mountable trays that house fans to blow air through the server rack and would be mounted where hot spots are detected.

The room that the servers are in will have to be kept cool as well. Often servers will be placed on a raised floor to have air conditioning run underneath. If there are multiple servers, they may be configured in a cold aisle hot aisle setting. This is where the front sides of the servers are turned to face each other, pulling in cold air from the air conditioner. The server's backside faces each other and expels heat out of the back. This ensures the servers are getting fresh cool air and safely expelling heat.

There is also liquid cooling for servers, but like a regular computer, this comes with a much higher price tag. One thing to keep in mind with the fans and cooling is the amount of noise that these fans will generate. Precautions to protect one's hearing should be taken.



Figure 3.3: Displaying a hot-swappable server fan.

Exercise 4 - BIOS and UEFI

BIOS and UEFI are firmware and allow the computer's hardware to function. In this exercise, you will learn about BIOS and UEFI.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- BIOS
- UEFI
- Firmware Updates

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

BIOS

Basic Input Output System (BIOS) is the firmware that allows the computer to perform its power on self-test, loads the hardware, and boots the system. This firmware is stored on a nonvolatile memory chip on the motherboard.

The BIOS menu will need to be accessed once the machine is switched on. Each manufacturer will have a slightly different way to access it, but it tells you when you see the company's splash screen. The enter key, ESC key, F1 and F8 are common options.

For security purposes, a BIOS password can be added. If someone attempts to access the BIOS and make changes, they will need to enter the password. Once into the BIOS menu, navigation must be done with the keyboard and monitor only. BIOS is on 8 or 16 bits, so there is no ability to add the functionality of the mouse. In the BIOS, you can adjust basic elements of how the computer functions. You can get an overview of the system, configure the CPU (i.e.) overclocking, configure hardware, remote access, manage virtualization, choosing the boot order, and more options.

UEFI

Unified Extensible Firmware Interface (UEFI) is essentially an update to BIOS. It uses 32 or 64 bits to operate and can give the user a graphical user interface experience, including a mouse. Some security enhancements prevent unverified applications from being loaded using Secure Boot. Overall, the performance is better when booting, and the drive size capabilities are vastly different. BIOS has a maximum drive size of 2 TB, and UEFI has a staggering 9 zettabytes maximum drive size. UEFI is the next generation, and some manufacturers have stopped supporting BIOS in favor of UEFI.

Firmware Updates

Firmware is a program that is written for hardware devices. These instructions give the device functionality and allow it to work with an operating system. The instructions are stored on a memory chip that is nonvolatile. These programs need to be updated occasionally, also known as flashing, to ensure that the firmware can make the hardware function properly and efficiently.

Sometimes additional functionality can be acquired by updating the firmware of a device. The update process can be as simple as following a few prompts to manually having to install the software. It is essential that during this process, electricity is constant, as any interruption can cause corruption. Both BIOS and UEFI are examples of firmware.

Review

Well done, you have completed the **Server Hardware Maintenance** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Local Hardware Administration
- Exercise 2 - Out-of-Band-Management
- Exercise 3 - Hot-Swappable Hardware
- Exercise 4 - BIOS and EUFI

You should now have further knowledge of:

- Keyboard Video Mouse (KVM)
- Crash Cart
- Serial Connectivity Vs Parallel Connectivity
- Console Connection
- Virtual Machine Administration Console
- Remote Drive Access
- Remote Console Access
- Remote Power On/Off
- Internet Protocol Keyboard Video Mouse (IP KVM)
- Hard Drives
- Network Attached Storage Cages
- Network Interface Cards (NIC)
- Power Supplies
- Fans and Cooling
- BIOS
- UEFI
- Firmware Updates

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.